Syrian Private University

Faculty of Artificial Intelligence Engineering

Senior Project 1 Report on

AI-Based Detection and Localization of Brain Aneurysm from CT scans

Submitted to

Department of Artificial Intelligence Engineering

Submitted by

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Under the Supervision of

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Faculty of Artificial Intelligence Engineering

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CERTIFICATE OF APPROVAL

SUPERVISOR CERTIFICATION

I certify that the preparation of this project prepared by Enas Saleh Intracranial Aneurysm Detection

AI-Based Detection and Localization of Brain Aneurysm/Bleeding from CT-Scans was made under my supervision at the Faculty of Engineering in partial fulfillment of the Requirements for the Degree of Bachelor

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Abstract

Brain aneurysm and intracranial hemorrhage are critical medical conditions that can lead to severe neurological complications, including stroke and death, if not diagnosed at an early stage. Computed Tomography (CT) imaging is widely used for brain diagnosis; however, manual interpretation is time-consuming and highly dependent on radiological expertise. This motivates the development of automated and reliable diagnostic support systems.

This study proposes an artificial intelligence-based framework for the classification of brain CT images into aneurysm-related cases, intracranial hemorrhage, and normal cases. The system employs Convolutional Neural Networks (CNNs) to automatically extract discriminative features for accurate classification. To enhance clinical applicability, a segmentation module is incorporated to localize abnormal regions associated with aneurysms and bleeding.

The model is trained and evaluated using publicly available datasets, including the "Brain Stroke CT Dataset" for hemorrhage and normal cases (also used for mask generation), and the "Computed Tomography (CT) of the Brain" dataset for aneurysm cases. To address the black-box nature of deep learning models, Explainable Artificial Intelligence (XAI) using Grad-CAM is applied to highlight regions influencing model predictions, improving interpretability and clinical trust. Data preprocessing and augmentation techniques are used to improve generalization.

Experimental results show strong performance, **with an accuracy of 92.5% and an F1-score of 92.4%** for classification. For segmentation, the model achieves a **Dice score of 0.65 and an IoU of 0.53**, indicating effective localization of abnormalities.

In conclusion, the proposed system integrates classification, segmentation, and explainability to provide a reliable and interpretable diagnostic tool for early detection of brain aneurysms and intracranial hemorrhage, supporting improved clinical decision-making.

الملخص

يُعد تمدد الأوعية الدموية الدماغية والنزيف داخل الدماغ من الحالات الطبية الحرجة التي قد تؤدي إلى مضاعفات عصبية خطيرة مثل السكتة الدماغية والوفاة إذا لم يتم تشخيصها مبكرًا تُستخدم صور الأشعة المقطعية (CT-Scan)

على نطاق واسع في تشخيص الدماغ إلا أن تفسيرها يدويًا يستغرق وقتًا طويلاً ويعتمد بشكل كبير على خبرة أطباء الأشعة مما يحفز الحاجة إلى أنظمة تشخيص آلية دقيقة وموثوقة

يقترح هذا البحث إطارًا يعتمد على الذكاء الاصطناعي لتصنيف صور الدماغ المقطعية إلى حالات تمدد الأوعية الدموية والنزيف الدماغي والحالات الطبيعية يعتمد النظام على الشبكات العصبية الالتفافية

لتحديد المناطق لاستخراج الخصائص المميزة تلقائيًا وتحقيق دقة عالية في التصنيف كما تم دمج وحدة للتجزئة المصابة المرتبطة بتمدد الأوعية الدموية أو النزيف

تم استخدام مجموعتي بيانات متاحتين للعامة حيث استُخدمت مجموعة

Brain Stroke CT Dataset

بينما استُخدمت مجموعة Masks لحالات النزيف والحالات الطبيعية بالإضافة إلى توليد الأقنعة

لحالات تمدد الأوعية الدموية Computed Tomography CT of the Brain

ولمعالجة مشكلة الصندوق الأسود في نماذج التعلم العميق تم استخدام تقنيات الذكاء الاصطناعي القابل للتفسير مثل GRAD-CAM, XAI

لتوضيح المناطق المؤثرة في قرارات النموذج مما يعزز الشفافية والثقة كما تم استخدام تقنيات المعالجة المسبقة وتوسيع البيانات لتحسين التعميم

أظهرت النتائج التجريبية أداءً قويًا حيث حقق النموذج دقة تصنيف بلغت 92.5 بالمئة ودرجة

F1-SCORE بلغت 92.4 بالمئة أما في مهمة التجزئة فقد حقق النموذج معامل

قدره IOU 0.53,0.65 Dice

مما يدل على قدرة جيدة في تحديد المناطق المصابة .

في الختام يقدم هذا النظام إطارًا فعالًا يجمع بين التصنيف والتجزئة وقابلية التفسير مما يساعد في دعم التشخيص المبكر لتمدد الأوعية الدموية الدماغية والنزيف الدماغي وتحسين القرارات السريرية

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Abbreviation	Full Term
AI	Artificial Intelligence
CNN	Convolutional Neural Network
CT	Computed Tomography
CTA	Computed Tomography Angiography
CAD	Computer-Aided Diagnosis
DL	Deep Learning
ML	Machine Learning
CV	Computer Vision
XAI	Explainable Artificial Intelligence
Grad-CAM	Gradient-weighted Class Activation Mapping
CAM	Class Activation Map
ReLU	Rectified Linear Unit
ROI	Region of Interest
UI	User Interface

IA	Intracranial Aneurysm
SAH	Subarachnoid Hemorrhage
ICH	Intracranial Hemorrhage
NN	Neural Network
DPI	Dots Per Inch
API	Application Programming Interface
GPU	Graphics Processing Unit
CPU	Central Processing Unit
RGB	Red Green Blue

Table 1 Abbreviation vs Full Term

Keywords

Brain Aneurysm, Computed Tomography (CT), Deep Learning, Convolutional Neural Networks (CNN), Medical Image Analysis, Image Classification, Image Localization, Grad-CAM, Explainable AI (XAI), Computer-Aided Diagnosis (CAD)

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Chapter 1: Introduction

Brain aneurysm is a serious and potentially life-threatening condition caused by the weakening of a blood vessel in the brain, leading to abnormal dilation or bulging [1]. If ruptured, it can result in severe complications such as stroke, brain damage, or intracranial bleeding [2]. Early detection is therefore critical to reducing mortality and improving patient outcomes.

Computed tomography (CT) scans are widely used for diagnosing brain aneurysms and bleeding due to their ability to provide detailed visualization of brain structures [3]. However, the interpretation of CT images is typically performed manually by radiologists, making the process time-consuming and highly dependent on expertise [4]. This can lead to variability in diagnosis and increase the risk of missed or delayed detection, particularly in subtle or early-stage cases.

Detecting aneurysms and hemorrhage remains challenging due to their small size, complex shapes, and similarity to surrounding anatomical structures [5]. Additionally, variations in image quality and the limited availability of annotated medical datasets further complicate the analysis. Traditional approaches often focus only on image classification, which does not provide sufficient information about the exact location of abnormalities or the reasoning behind predictions [6].

This project aims to develop an AI-based system for the detection and localization of brain aneurysms and hemorrhage from CT scans. The proposed approach leverages deep learning techniques to identify abnormalities and highlight the regions of interest within medical images. Furthermore, it incorporates interpretability methods to provide insights into the model's decision-making process, enhancing transparency and trust. The ultimate goal is to support clinicians with a more accurate, efficient, and explainable diagnostic tool.

Chapter 2: Project Description

2.1 Background

Brain aneurysm is a life-threatening condition caused by the weakening of blood vessels in the brain, affecting approximately **3–5% of the global population [5]**. If ruptured, it can lead to intracranial hemorrhage, with a mortality rate reaching **up to 50% [6]**, and many survivors suffering long-term neurological complications.

Computed Tomography (CT) imaging is widely used for diagnosis due to its speed and effectiveness in detecting brain abnormalities [3]. However, identifying aneurysms and hemorrhage in brain CT scans remains challenging and highly dependent on radiologist expertise [4]. Recent advances in deep learning, particularly Convolutional Neural Networks (CNNs), offer promising solutions to improve detection accuracy and support clinical decision-making [5].

2.2 Problem Statement

The detection of brain aneurysms from CT images remains a complex and critical challenge in medical diagnostics. Despite the availability of high-resolution imaging, accurately identifying aneurysms is difficult due to their subtle appearance and similarity to surrounding tissues. Small or unruptured aneurysms are particularly challenging to detect, increasing the risk of misdiagnosis or delayed intervention.

The diagnostic process relies heavily on manual interpretation by radiologists, which is time-consuming and subject to variability based on experience and workload. This can lead to inconsistencies in diagnosis and may affect the reliability of clinical decisions, especially in high-pressure healthcare environments.

Another significant challenge is the limited availability of large, well-annotated medical datasets. This restricts the ability of computational models to generalize effectively across diverse cases, reducing their performance in real-world applications.

Furthermore, many traditional approaches focus only on classification without providing precise localization of abnormalities or explaining the reasoning behind predictions. This lack of interpretability reduces trust in automated systems and limits their practical use in clinical settings.

Therefore, there is a need for an intelligent system that improves detection accuracy while providing reliable localization and interpretable results. Such a system can support healthcare professionals in making faster, more consistent, and informed diagnostic decisions, ultimately improving patient outcomes.

2.3 Project Scope

This project focuses on the detection, Explaining and localization of brain aneurysms using computed tomography (CT) brain images. It emphasizes the identification of vascular abnormalities, particularly abnormal dilation of blood vessels that may indicate the presence of an aneurysm.

In addition to aneurysm detection, the project considers associated indicators such as intracranial bleeding, which is a critical condition often resulting from aneurysm rupture. By addressing both the primary condition and its potential consequences, the study aims to provide a more comprehensive analysis.

The project utilizes a dataset of CT brain scans that includes cases of aneurysm, bleeding, and normal conditions. Data preprocessing and augmentation techniques are applied to enhance dataset representation and improve the robustness and generalization of the model.

The system is designed as a supportive diagnostic tool that assists in identifying abnormalities and highlighting the affected regions within the images. It also incorporates interpretability techniques, such as visualization methods, to improve understanding and confidence in the results from AI.

However, this project does not aim to replace clinical judgment or provide a complete medical diagnosis. It is intended solely as a decision-support system for healthcare professionals. Additionally, the scope is limited to CT imaging data and does not include other modalities such as MRI or real-time clinical deployment

2.4 Project objective

The main objective of this research is to improve the detection and localization of brain aneurysms in computed tomography (CT) images, with the aim of supporting more accurate, reliable, and efficient diagnosis.

To achieve this goal, the study focuses on the following objectives:

- To develop a robust approach capable of accurately identifying the presence of brain aneurysms in medical images.
- To enhance the ability to precisely localize affected regions within the images, providing clearer visual guidance for analysis.
- To improve data representation through preprocessing and augmentation techniques in order to address the limitations of small medical datasets.
- To provide more interpretable results by incorporating explainability methods that offer deeper insight into the factors influencing the model's decisions.
- To support faster and more consistent diagnostic processes, thereby reducing reliance on manual interpretation while assisting clinical decision-making.

2.5 Project Features

The proposed system for detecting and localizing brain aneurysms and bleeding from CT images includes several key features designed to enhance accuracy, efficiency, and usability in medical image analysis.

Automated Detection:
The system automatically analyzes CT brain images to detect abnormalities such as aneurysms and intracranial bleeding without manual intervention.

Accurate Localization:
The model highlights regions of interest within the images, enabling precise identification of affected areas and supporting better clinical analysis.

Deep Learning-Based Approach:

The system employs Convolutional Neural Networks (CNNs) via a dual-stage architecture for automated aneurysm detection. By utilizing ResNet50 for feature extraction and an Attention U-Net for pixel-level segmentation, the model identifies complex patterns in brain CT and MRI scans.

Data Preprocessing and Augmentation:

Preprocessing and augmentation techniques are applied to enhance image quality and increase dataset diversity, improving model generalization.

Explainable AI (XAI):

The system incorporates interpretability methods such as Grad-CAM++ to provide visual explanations of model predictions.

User Interface (UI):

The system includes a user-friendly interface that allows users to upload CT images and visualize model predictions along with highlighted regions of interest. This feature demonstrates the practical applicability of the model as a decision-support tool.

Performance Evaluation:

The model is evaluated using multiple metrics, including accuracy, precision, recall, F1-score, IuO and DICE ensuring reliable assessment.

Efficiency and Scalability:

The system is capable of processing multiple images efficiently, making it suitable for real-world applications.

2.6 Project Feasibility

The feasibility of the proposed system for detecting and localizing brain aneurysms and bleeding using CT images is evaluated from multiple perspectives to ensure that the project can be successfully developed and implemented within the available resources and constraints.

2.7 System Requirements

The implementation of the proposed deep learning-based system for detecting and localizing brain aneurysms from CT images requires suitable computational resources and software tools to support data processing, model training, evaluation, and prediction.

2.8 Beneficiaries of the Research

The outcomes of this research can benefit multiple stakeholders across the healthcare and technology sectors. Healthcare professionals, including radiologists and neurologists, can use the proposed system as a decision-support tool to improve the accuracy and efficiency of aneurysm detection while reducing diagnostic workload.

Patients are indirect beneficiaries, as early and accurate detection can lead to timely medical intervention and improved health outcomes. Hospitals and medical institutions can also benefit by enhancing diagnostic efficiency and reducing the time required for image analysis.

In addition, medical researchers and academics can use this work as a foundation for further studies in medical image analysis and artificial intelligence. AI developers may also leverage the proposed methodology to build more advanced diagnostic systems. Furthermore, the system can serve as a learning tool for medical students, helping them better understand CT imaging and AI-assisted diagnosis.

Chapter 3: Theoretical Study

3.1 Introduction

This chapter presents the theoretical and computational foundations related to the proposed project. It explains the main concepts used in developing an AI-based system for detecting and localizing brain aneurysms from CT scan images. The chapter focuses on the relevant theories and techniques applied in the project, including medical image processing, deep learning, convolutional neural networks, transfer learning, image classification, Grad-CAM-based localization, mask generation, attention-based segmentation, loss functions, optimization methods, and evaluation metrics.

The purpose of this chapter is to provide the scientific background required to understand how the proposed system works. Only concepts directly related to the implementation of the project are discussed, while unrelated theoretical topics are avoided. The chapter also supports the study by referring to established methods and previous research in medical image analysis and deep learning.

3.2 Computed Tomography (CT) Brain Imaging

Computed Tomography (CT) imaging is a widely used medical imaging technique that provides detailed cross-sectional views of internal body structures, including the brain. It is particularly effective in detecting abnormalities such as hemorrhage, tumors, and vascular conditions due to its high spatial resolution and fast acquisition time.

In the context of brain aneurysm detection, CT scans play a critical role in identifying structural changes in blood vessels, such as abnormal dilation or rupture. CT angiography (CTA), a specialized form of CT imaging, is commonly used to visualize blood vessels more clearly and assist in detecting aneurysms.

However, CT images present several challenges for automated analysis. Aneurysms are often small and can appear similar to surrounding tissues, making them difficult to distinguish. Additionally, variations in image quality, noise, and patient-specific anatomical differences further complicate the detection process.

To address these challenges, CT images typically undergo preprocessing steps before being used in deep learning models. These steps may include normalization, resizing, and contrast enhancement to improve the quality and consistency of the input data.

In this project, CT brain images serve as the primary data source for training and evaluating the proposed deep learning model. The system relies on these images to learn patterns associated with aneurysms and to identify and localize abnormal regions effectively.

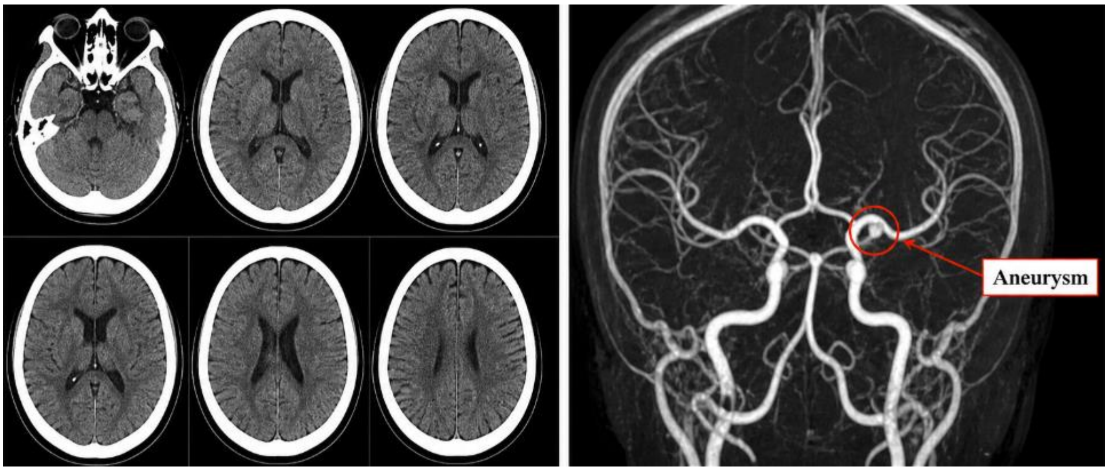

Figure 1: CT and CTA Brain Imaging Examples

Figure 3.1: CT and CTA Brain Imaging Examples

3.3 Medical Image Preprocessing

Preprocessing is an essential step in preparing CT images for deep learning models. Raw medical images often contain noise and inconsistencies that may affect model performance.

Common preprocessing techniques include:

- Image resizing to a fixed resolution
- Normalization of pixel intensity values

- Contrast enhancement
- Noise reduction

In addition, data augmentation techniques such as rotation, flipping, and brightness adjustments are applied to improve dataset diversity and reduce overfitting.

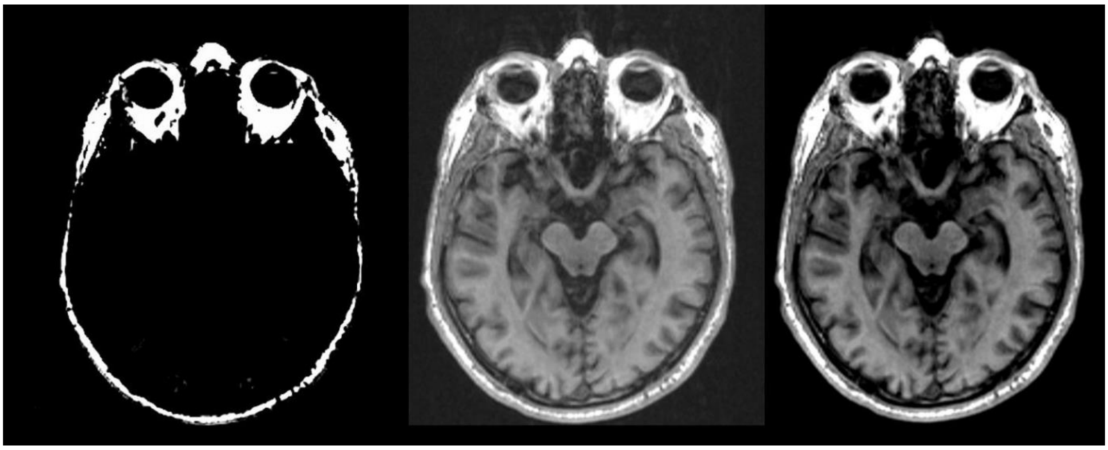

Figure 3.2: Data Augmentation Techniques Applied to CT Brain Images

3.4 Deep Learning and Convolutional Neural Networks (CNNs)

Deep learning enables models to automatically learn complex patterns from data. Convolutional Neural Networks (CNNs) are particularly effective for image analysis tasks.

A CNN consists of:

- Convolutional layers for feature extraction
- Activation functions (ReLU)
- Pooling layers for dimensionality reduction
- Fully connected layers for classification

In this project, CNNs are used to analyze CT images and detect patterns associated with aneurysms.

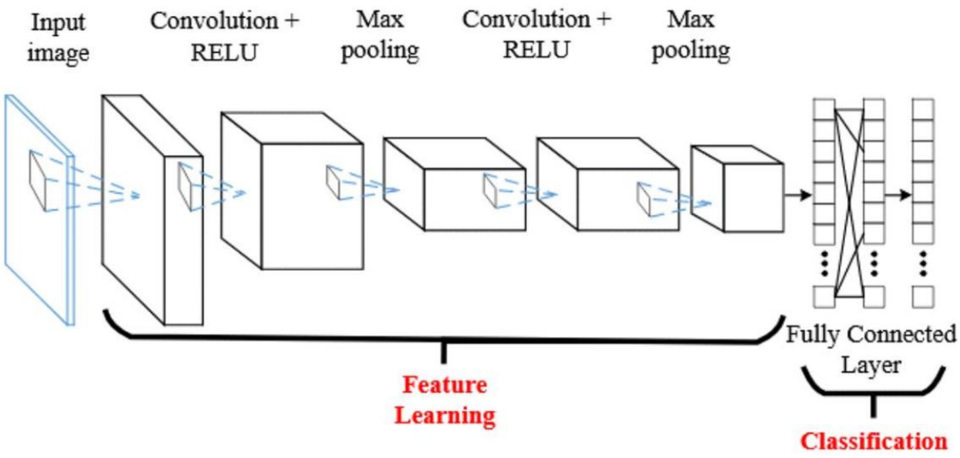

Figure 3.3: Convolutional Neural Network (CNN) Architecture

3.5 Image Classification and Explainability

Image classification assigns a label to an image (e.g., normal or abnormal (Bleeding and aneurysm), while localization identifies the regions where model made the predication occur.

In this project:

• Classification: Utilized to determine the presence of an aneurysm using the ResNet50 backbone.

• Explainability: Achieved through Grad-CAM++ (Gradient-weighted Class Activation Mapping), which generates heatmaps that highlight regions influencing the model's predictions. This addresses the challenge of AI interpretability by providing visual evidence for the classification result.

General Concepts in Deep Learning

ResNet50 (Residual Network)

ResNet50is a type of Convolutional Neural Network (CNN) architecture designed to handle extremely deep models without losing accuracy. Its breakthrough feature is the "skip connection," (F(x) = H(x) - x). which allows information to bypass certain layers. This solves the "vanishing gradient" problem, enabling the network to learn intricate patterns and textures that are essential for accurate image recognition.

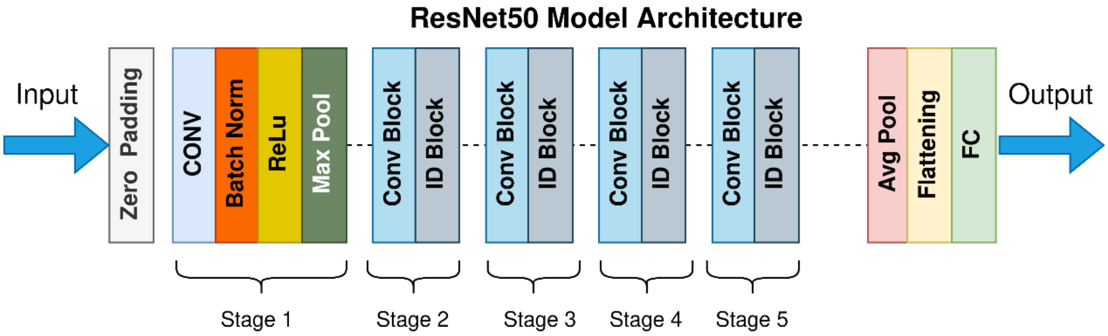

Figure 3.4 ResNet50is Architecture

Grad-CAM (Gradient-weighted Class Activation Mapping)

Grad-CAM is a visualization tool that explains the "why" behind an AI's decision. It analyses the gradients flowing into the final convolutional layer of a network like ResNet to identify which parts of an image are most important. The result is a heatmap overlaid on the original image, highlighting the specific regions the model focused on to reach its conclusion, which builds trust and transparency.

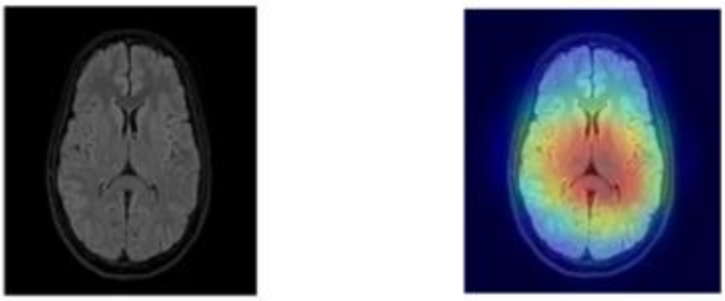

Figure 3.5: Grad-CAM Visualization Highlighting Regions of Interest

3.6 Model Training and Optimization

Model training is the process where the network learns to map input medical images to their corresponding labels by iteratively minimizing a Loss Function. This optimization ensures the model captures the intricate features of brain abnormal region while maintaining high generalizability.

Key Components of the Training Pipeline

Here are the keywords for your model training and optimization section:

- Loss Functions: Binary Cross-Entropy (BCE) with Logits (Classification), Combined Dice + BCE Loss (Segmentation).
- Optimizer: Adam (Adaptive Moment Estimation).
- Fine-tuning: Dual-stage training, ResNet50 weight freezing, Layer unfreezing.
- Learning Rate: Adaptive adjustment, Learning Rate Scheduler, Plateau reduction.
- Regularization: Early Stopping (Validation loss monitoring).

3.7 Evaluation Metrics

The model performance is evaluated using:

- **Accuracy:** Overall correctness
- **Precision:** Correctness of positive predictions
- **Recall (Sensitivity):** Ability to detect abnormal cases
- **F1-Score:** Balance between precision and recall

In medical applications, recall is particularly important to ensure that critical conditions are not missed.

For the segmentation component of this system, additional metrics are used to evaluate the quality of predicted masks against ground-truth annotations:

- **Dice Coefficient (Dice Score):** Measures the overlap between the predicted mask and the ground-truth mask. It is computed as $2 \times |A \cap B| / (|A| + |B|)$, where A is the predicted mask and B is the ground truth. A Dice score of 1 indicates a perfect overlap. It is particularly suited to medical segmentation tasks where the lesion region is small relative to the image background.
- **Intersection over Union (IoU / Jaccard Index):** Evaluates the ratio of the intersection area to the union area between the predicted and ground-truth masks: $IoU = |A \cap B| / |A \cup B|$. IoU is a stricter metric than the Dice coefficient and is widely used in semantic segmentation benchmarks to quantify spatial accuracy of predicted regions.

Binary Cross-Entropy Loss (BCE): is the standard objective function for binary classification. It measures the difference between the actual label and the predicted probability for a specific class.

3.8 Image Segmentation

Image segmentation is the task of assigning a class label to every pixel in an image, thereby delineating the exact spatial boundaries of regions of interest. In the medical imaging context, segmentation allows a model to not only detect the presence of a pathology but also to precisely delineate which pixels correspond to abnormal tissue. This is critical for clinical applications such as measuring lesion volume, computing anatomical metrics, and enabling surgical planning.

3.8.1 U-Net Architecture

The U-Net architecture (Ronneberger et al., 2015) is the foundational encoder-decoder network for medical image segmentation. It features a symmetric structure: a contracting encoder path that progressively reduces spatial resolution while increasing feature depth (using convolution and max-pooling), and an expansive decoder path that restores spatial resolution (using up-sampling and transposed convolutions). Skip connections between corresponding encoder and decoder layers allow the network to combine high-level semantic information from the bottleneck with fine-grained spatial detail from earlier layers, enabling precise pixel-level predictions even when only a limited number of training samples are available.

3.8.2 Attention Gates in Segmentation

Attention gates (Oktay et al., 2018) extend the standard U-Net by learning to emphasize relevant spatial regions and suppress irrelevant background activations during decoding. An attention gate receives two inputs: a gating signal from the deeper decoder layer (which carries semantic context) and the skip-connection feature map from the encoder. It computes a soft spatial attention coefficient between 0 and 1 for each position, which is multiplied element-wise with the skip-connection features before merging. This mechanism allows the network to focus on lesion-relevant areas of the CT scan without requiring additional supervision, which is particularly beneficial when training data is limited.

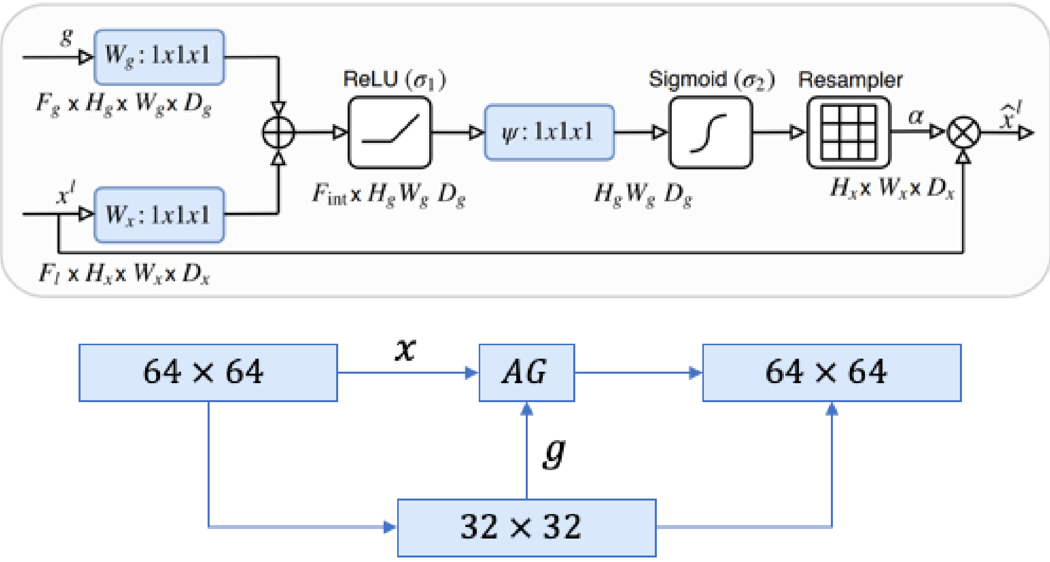

Figure 3.6 Unet with Attention Architecture

3.8.3 Segmentation Loss Functions: Dice and BCE

Training a binary segmentation model on medical images requires a loss function that handles the severe class imbalance between lesion pixels (minority) and background pixels (majority). Two complementary losses are used in this project. The Dice Loss directly optimizes the Dice coefficient by maximizing the overlap between predicted and ground-truth masks. It is defined as: L_Dice = 1 − (2 × |P ∩ G| + ε) / (|P| + |G| + ε), where P is the predicted probability map, G is the ground-truth mask, and ε is a small smoothing constant. The Binary Cross-Entropy with Logits Loss (BCEWithLogitsLoss) evaluates each pixel independently as a binary classification problem. When combined with equal weighting (0.5 × Dice + 0.5 × BCE), the resulting DiceBCE loss is more stable than using either term alone, particularly when training on noisy or limited annotations, and has become the standard choice for medical image segmentation tasks.

3.8.4 Ground-Truth Mask Generation from Pixel-Difference Pairs

Because manually annotated segmentation masks for brain CT datasets are scarce, the segmentation component of this project derives binary masks computationally from paired image data available in the Brain Stroke CT Dataset. Each bleeding case provides two aligned images: a plain CT scan (PNG) and an overlay version (OVERLAY) in which the hemorrhagic region is highlighted. By computing the per-pixel absolute difference between these two images in the RGB color space and applying a threshold, a binary mask is extracted that marks the lesion boundary. This pixel-difference masking approach enables supervised segmentation training without the need for time-consuming manual annotation, while still providing spatially accurate ground-truth labels derived from the original radiological markings.

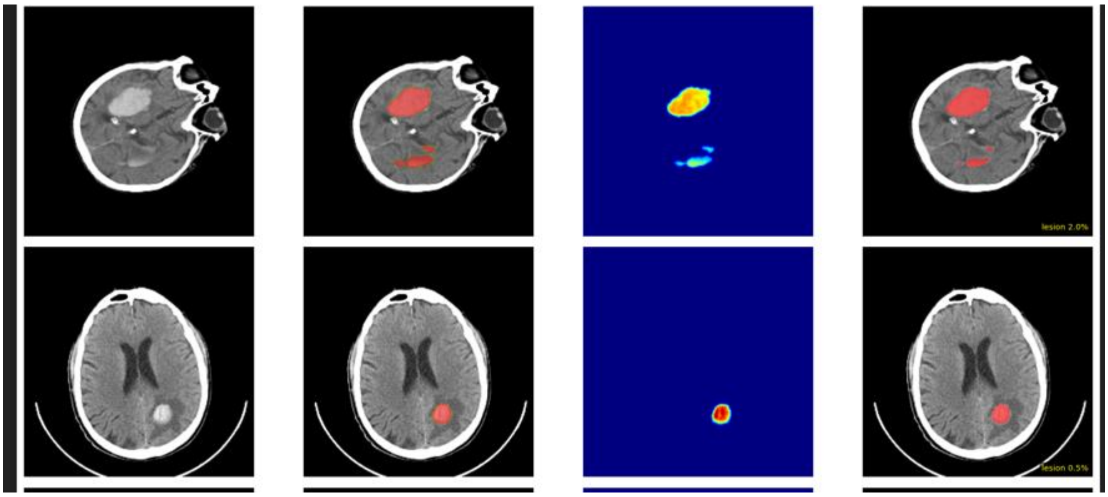

Figure 3.7: PNG and OVERLAY with the difference between the two images

Chapter 4: Literature Review

4.1 introduction

In the first phase of this project, we built a deep learning model for binary classification of CT brain scans — determining whether an intracranial aneurysm (IA) and hemorrhage is present or absent.

However, classification alone answers only the question "Is there an issue?" It does not tell clinicians where in the brain damage is located, which cerebrovascular territory is affected, or how large the lesion is.

The current phase of the project therefore extends the pipeline in two directions. First, Explainability: after classifying a scan as positive, we apply Gradient-weighted Class Activation Mapping (Grad-CAM) to highlight the spatial region within the CT scan driving the prediction, enabling the model to function as an attentional guide for the reporting radiologist [5, 6]. Second, segmentation: for confirmed positive cases, a dedicated segmentation module (based on U-Net with Attention) delineates the precise voxel-level boundary of the aneurysm and damage vessel in the brain hemorrhage, enabling automated measurement of diameter, neck width, and aspect ratio [2, 3].

The six papers reviewed below were selected to provide the evidence base for each stage of this extended pipeline. They cover: (a) 2D classification as the established starting point [1]; (b) state-of-the-art multicenter CTA detection and segmentation that represent the clinical target [2, 3, 4]; and (c) explainable AI methods using Grad-CAM that bridge classification and spatial localization without requiring additional pixel-level annotation [5, 6].

4.2Related-Works

Table 1 below summarizes the six selected papers across seven review dimensions. Papers are ordered to reflect the progression from pure classification [1] → combined detection and segmentation [2, 3, 4] → explainable AI localization [5, 6]. All six papers are peer-reviewed and published in indexed journals (Sensors, Radiology, Scientific Reports, Nature Communications, BMC Medical Imaging, Biomedical Signal Processing and Control).

[1] Hlavata et al. — Automated Method for Intracranial Aneurysm Classification Using Deep Learning Sensors, 2024	
Dataset Used	DiscoverAI / Roboflow Universe — 611 annotated CT images (tumour, cancer, aneurysm); 87/8/5 train/val/test split
Training Approach	Supervised 2D CNN; Adamax optimiser; LR = 0.001; 50 epochs; batch = 16; benchmarked vs ResNet-50/101/152 and VGG-16
AI Model	Custom 2D CNN (3 conv + pooling + dropout + dense + softmax); compared with ResNet-50, ResNet-101, ResNet-152, VGG-16
Feature Extraction	Automatic hierarchical feature learning from 2D axial CT slices; no handcrafted feature engineering
Evaluation & Results	Proposed CNN: 98.01% accuracy, loss = 0.036; ResNet-152: 98.0% (best baseline); compact model = faster inference — good for screening entry point
Year	2024

Figure 4.1: Related Work — Study 1

[2] Wei, Song et al. — Knowledge-Augmented Deep Learning for Segmenting and Detecting Cerebral Aneurysms at CTA: A Multicenter Study *Radiology*, 2024	
Dataset Used	6,060 CTA patients from 8 hospitals (Feb 2018–Oct 2021); external DSA test set: 118 patients (1 hospital, Feb–Feb 2023); split: 4,342 train / 1,086 val / 632 test
Training Approach	Supervised encoder-decoder 3D CNN; anatomical knowledge augmentation; internal + external DSA validation; DeLong test for AUC comparison vs radiologist reports
AI Model	Knowledge-augmented 3D CNN (nnU-Net-based encoder-decoder); integrates vessel anatomical priors as auxiliary inputs during training
Feature Extraction	3D volumetric CTA features; anatomical vessel topology (circle of Willis priors); multi-scale hierarchical encoder-decoder feature maps
Evaluation & Results	DSC = 0.87 (segmentation); Sensitivity = 85.7% per vessel; AUC = 0.93 vs radiologist: 0.91 (p = 0.67, non-significant); Processing time: 1.76 ± 0.32 min/scan
Year	2024

Figure 4.2: Related Work — Study 2

[3] Ham, Seo et al. — Automated Detection of Intracranial Aneurysms Using Skeleton-based 3D Patches, Semantic Segmentation, and Auxiliary Classification *Scientific Reports*, 2023	
Dataset Used	154 TOF-MRA scans (internal, Seoul Nat'l Univ. Bundang Hospital); 113 ADAM challenge scans (external validation); varied positive/negative patch ratios 1:1 to 5:1
Training Approach	Multi-task learning: simultaneous segmentation + auxiliary classification; skeleton-based 3D patch sampling to overcome class imbalance; internal + external cross-site validation
AI Model	3D U-Net with auxiliary classification branch (multi-task); skeleton-based vessel-patch extraction strategy for balanced training
Feature Extraction	Automatic 3D spatial features from vessel-skeleton-derived patches; multi-task gradient sharing between segmentation and classification heads
Evaluation & Results	Internal accuracy = 0.910; external accuracy = 0.883; Dice = 0.755; Sensitivity = 0.882; ~0.3 FP/case — robust across scanner vendors
Year	2023

Figure 4.3: Related Work — Study 3

[4] Shi, Miao, Schoepf et al. — A Clinically Applicable Deep-Learning Model for Detecting Intracranial Aneurysm in CTA Images *Nature Communications*, 2020	
Dataset Used	Multiple multicenter CTA cohorts (8 hospitals); diverse CT scanner manufacturers and image quality levels; large-scale multicenter dataset with radiologist & neurosurgeon benchmarks
Training Approach	End-to-end 3D CNN; dual attention mechanism; residual connections; 8-cohort cross-validation; benchmarked against radiologists and neurosurgeons at patient-level sensitivity
AI Model	3D CNN encoder-decoder with dual attention (spatial + channel); residual blocks; end-to-end detection pipeline
Feature Extraction	Long-range contextual 3D features via dual spatial-channel attention; vessel bifurcation residual feature maps; whole-volume 3D spatial context
Evaluation & Results	Patient-level sensitivity surpasses both radiologists and neurosurgeons across all 8 cohorts; clinically validated; robust to scanner and image quality variation
Year	2020

Figure 4.4: Related Work — Study 4

[5] Musthafa, Mahech et al. — Enhancing Brain Tumor Detection in MRI Images Through Explainable AI Using Grad-CAM with ResNet-50 *BMC Medical Imaging*, 2024	
Dataset Used	BR35H brain MRI dataset (Kaggle) — ~3,060 images (tumorous / non-tumorous); data augmentation applied; training set ~80%; split: 80/10/10 (train/val/test)
Training Approach	Transfer learning: ResNet-50 fine-tuned on brain MRI; two-stage training (frozen backbone → full fine-tune); optimizer: PyTorch; Grad-CAM applied post-classification; final conv layer for spatial localisation
AI Model	ResNet-50 (ImageNet pretrained → fine-tuned); Grad-CAM for post-classification spatial heatmap generation; SHAP and LIME as secondary XAI methods
Feature Extraction	Deep convolutional feature maps from ResNet-50 final layers; gradient-weighted class activation maps (Grad-CAM) isolate the spatial region driving classification — acts as localisation without pixel-level annotation
Evaluation & Results	Training accuracy = 100%; Validation accuracy = 98.67%; Precision, Recall, F1 = 98.50%; Grad-CAM heatmaps confirmed to spatially co-localise with radiologist-annotated lesion regions
Year	2024

Figure 4.5: Related Work — Study 5

[6] Solak et al. — Explainable Deep Learning Framework for Brain Tumor *Information Sciences*, 2025	
Dataset Used	BRATS2019 dataset (Kaggle) — ~3,000 brain MRI images
Training Approach	Two-stage CNN training: Stage 1: standard training (97.2% acc); Stage 2: XAI-mask augmentation (Grad-CAM + LIME + SHAP outputs used as additional supervision)
AI Model	Custom CNN with two-stage training; Grad-CAM (spatial heatmap)
Feature Extraction	Grad-CAM: highlights broad but clinically relevant tumour regions
Evaluation & Results	Stage 1: Accuracy = 97.20%, Sensitivity = 98.00%, Specificity = 96.40%, AUC = — Stage 2 (XAI-augmented): Accuracy = 99.40%, Sensitivity = 99.20%, Specificity = 99.60%, AUC = 99.50%
Year	2025

Figure 4.6: Related Work — Study 6

The project adopts a tiered approach influenced by [1] establishes a fast 2D entry point, ensuring the system is computationally practical before engaging intensive 3D modules.

Clinical Precision: [2, 3, 4] justify using anatomical priors, attention gates, and multi-task learning. These influences ensure the segmentation module is robust across different scanners and precise enough for surgical planning.

Transparency & Refinement: [5, 6] introduce Grad-CAM to transform the "black box" into a visual guide. By using XAI heatmaps as a feedback loop, the project achieves higher accuracy while providing the explainability clinicians require.

4.3 Research gap

Despite the promising results achieved by previous studies, several limitations still exist. Many approaches focus primarily on classification without providing precise localization of abnormalities. On the other hand, segmentation-based methods require large, well-annotated datasets, which are often difficult to obtain in medical domains.

Moreover, the complexity of advanced models such as 3D CNNs and multi-task architectures may limit their practical deployment due to high computational cost. In addition, although interpretability methods have been introduced, they are not consistently integrated into the detection pipeline.

Therefore, there remains a need for a balanced approach that combines accurate classification, effective localization, and interpretability while maintaining computational efficiency and robustness to limited data.

Chapter 5: Methodology

5.1 Introduction

This chapter presents the methodology used to develop the proposed system for detecting and localizing brain aneurysms from CT images. It describes the overall workflow of the system, including data collection, preprocessing, model design, training process, and evaluation procedures.

The chapter also explains the techniques and tools used to implement the deep learning model, as well as the pipeline followed to ensure accurate and reliable results.

5.2 System Overview

The proposed system is designed to automatically detect and localize brain aneurysms and hemorrhage from CT images using deep learning techniques. The system consists of several main stages that work together to process input images and generate meaningful predictions.

The overall workflow of the system can be summarized as follows:

Here is the high-level overview of the proposed system, organized by its functional stages:

1. Input

- **Data Ingestion**: Raw 2D CT scans.

2. Classification & Feature Extraction

- Backbone: CNN (ResNet50).

- Binary Prediction: Identification of Presence vs. Absence of aneurysms or hemorrhages.

3. Explainable Localization (XAI)

- Technique: Grad-CAM.
- Visual Transparency: Generation of gradient-weighted heatmaps.
- Attentional Guide: Highlighting specific spatial regions driving the classification to build clinical trust.

4. Attention-Guided Segmentation

- **Architecture**: **Attention U-Net**.

5. Output

- Prediction
- Visual Explanation: Combined diagnostic scan and heatmap.
- Clinical Utility: Final report generation for targeted surgical planning and monitoring.

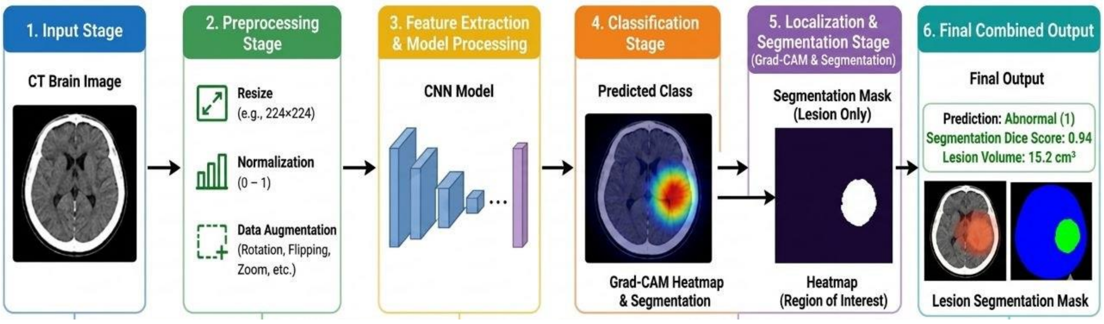

Figure 5.1: Full system pipeline

5.3 Dataset Description

This study utilizes publicly available datasets of brain CT images collected from Kaggle. The datasets include a variety of cases representing normal conditions, brain abnormalities, and stroke-related findings, which are relevant to the detection of brain aneurysms and associated conditions(hemorrhage).

The primary datasets used in this research are:

- Brain Stroke CT Dataset
- Computed Tomography (CT) Brain Dataset

These datasets provide a diverse collection of CT brain images that support the training and evaluation of the proposed deep learning model.

5.3.1 Dataset Characteristics

The datasets consist of CT brain images categorized into different classes, including

- Normal cases
- Abnormal cases (e.g, hemorrhage, or suspected aneurysm-related conditions)

The images vary in resolution, quality, and anatomical representation, reflecting real-world clinical variability.

5.3.2 Data Distribution

The dataset is divided into three main subsets:

- Training set
- Validation set
- Test set

  This division ensures that the model is trained effectively while also being evaluated on unseen data to measure its generalization performance.

5.3.2 Data Preprocessing

Before training, the images undergo several preprocessing steps:

- Resizing images to a fixed input size
- Normalizing pixel values
- Applying data augmentation techniques such as rotation and flipping

5.3.3 Data Augmentation

The implementation utilizes the (albumentations) library. The following augmentations were applied to increase model robustness:

- Geometric: Horizontal and vertical flips.
- Photometric: Random brightness and contrast adjustments to simulate varying CT scanner intensities.
- Spatial: Shift, scale, and rotate transformations.

These steps improve model robustness and help address the limitations of limited dataset size.

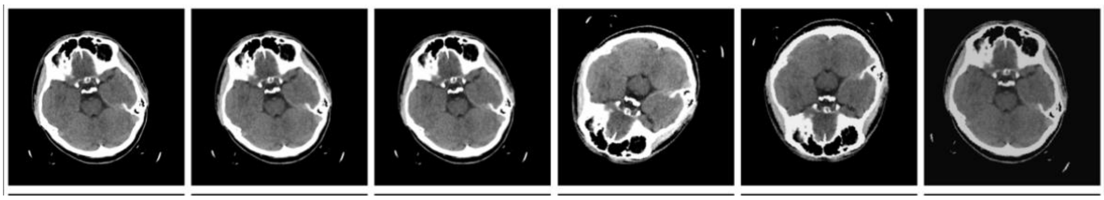

Figure 5.2: Image Preprocessing Pipeline (Normalization, Resizing and Augmentation*)*

5.4 Proposed Model Architecture

The proposed system implements a two-stage deep learning pipeline: a binary classification model followed by a dedicated segmentation model. Both models are trained independently and then combined in a full inference pipeline together with a Grad-CAM++ visualization module. All models operate on RGB CT images resized to 224×224 pixels, normalized using ImageNet mean and standard deviation values.

5.4.1 Model Structure

The architecture consists of multiple layers organized sequentially to extract features and perform classification. The main components of the model include:

- Convolutional Layers:
  These layers are responsible for extracting low-level and high-level features such as edges, textures, and shapes from CT images.

- Activation Function (ReLU):
  The Rectified Linear Unit (ReLU) introduces non-linearity into the model, allowing it to learn complex patterns.

- Pooling Layers:
  Pooling reduces the spatial dimensions of feature maps, helping to decrease computational complexity and prevent overfitting.

- Fully Connected Layers:
  These layers combine extracted features to perform the final classification.

- Output Layer (Softmax / Sigmoid):
  Produces the final prediction, indicating whether the image is normal or abnormal.

5.4.2 Feature Extraction

The classification model employs ResNet50 pretrained in Pytorch on ImageNet (IMAGENET1K_V2 weights) as the feature extraction backbone. ResNet50 is a 50-layer residual network that uses skip connections (residual connections) to avoid vanishing gradient problems during training. Transfer learning is applied in a two-stage strategy:

in Stage 1 .Freezing the initial layers to stabilize, all encoder layers except layer4 and the final fully connected head are frozen, and only the classification head is trained for an initial set of epochs;

in Stage 2, all parameters are unfrozen and fine-tuned end-to-end with a lower learning rate. The custom classification head replaces ResNet50's original fully-connected layer with: Dropout(0.4) → Linear(2048, 256) → ReLU → Dropout(0.3) → Linear(256, 1). The single output logit is passed through a sigmoid function to produce a binary probability score (positive = hemorrhage/aneurysm, negative = normal).

5.4.3 Classification Mechanism

The extracted features are passed through fully connected layers, where the model assigns a probability score to each class. The final classification is determined based on the highest probability value.

5.4.4 Explainabilty using Grad-CAM

The system implements Grad-CAM++ (Chattopadhay et al., 2018), which is an improved version of the original Grad-CAM. While standard Grad-CAM computes first-order gradients of the class score with respect to the final convolutional feature maps, Grad-CAM++ additionally uses second-order Taylor expansion coefficients (alpha weights) to produce more accurate localization, especially when multiple instances of the target class appear in an image. In this pipeline, Grad-CAM++ uses the final convolutional layer (layer4[-1]) of the ResNet50 as the target layer and an auxiliary layer for combined CAM computation. Four-way test-time augmentation (TTA) is applied: the original image, horizontal flip, vertical flip, and both flips are processed separately, and their heatmaps are averaged to produce a more robust and stable visualization. The resulting heatmap is applied only to images classified as positive (hemorrhage/aneurysm), and is used exclusively for visualization and interpretability; it is never converted to training masks or used as segmentation supervision.

This allows the system to provide visual explanations, making it easier to understand how the model reaches its decisions.

5.4.5 Model Advantages

The proposed architecture offers several advantages:

- Ability to highlight important regions using Grad-CAM
- Balanced performance between accuracy and computational cost

5.4.6 Segmentation Model: Attention U-Net

A custom Attention U-Net is implemented from scratch in PyTorch as the segmentation model.

- Encoder (Down-sampling): Four stages utilizing ConvBlock sequences (Conv2d, BatchNorm, ReLU, Dropout) that increase filter depth from 3 → 1024 to capture complex vascular features.

- Attention Gates: Integrated into skip connections. They use gating signals from deeper layers to selectively weight features, highlighting aneurysm pixels while suppressing irrelevant background noise.

- Decoder (Up-sampling): Uses bilinear up-sampling combined with the AttentionGate modules to reconstruct the spatial resolution of the mask.

- Output: A single-channel sigmoid-activated map representing the binary pixel-level segmentation.

5.4.7 Full Inference Pipeline

At inference time, both models and the Grad-CAM++ module are loaded and combined into a single pipeline (run_inference function). Given an input CT image path, the pipeline executes the following ordered steps: (1) Load and preprocess the image: convert to RGB, resize to 224×224, apply ImageNet normalization. (2) Classification: pass through the ResNet50 classifier; apply sigmoid to the single output logit; threshold at 0.5 to obtain a binary prediction and a confidence score. (3) Grad-CAM++ visualization (positive cases only): generate and overlay the attention heatmap on the original image to highlight the most relevant spatial regions driving the classification. (4) Segmentation: pass the preprocessed image through the Attention U-Net; apply sigmoid and a 0.5 threshold to produce a binary mask; apply morphological post-processing. (5) Output: return the classification label, confidence score, Grad-CAM++ heatmap image, and the segmentation mask. This end-to-end pipeline is designed for deployment on Google Colab with TorchScript-exported models.

5.5 Training Process

This section details the dual-stage training procedure designed to optimize both the classification and segmentation performance of the proposed system. The training process involves defining the loss function, selecting an optimization algorithm, and tuning hyperparameters to achieve the best performance.

5.5.1 Loss Function

Classification Stage: Cross-Entropy Loss is utilized to measure the divergence between predicted probabilities and true labels, facilitating accurate identification of abnormalities.

Segmentation Stage: A hybrid Dice + Binary Cross-Entropy (BCE) loss (50/50 split) is implemented. This combination effectively manages the severe class imbalance inherent in medical imaging, where the lesion pixels occupy a very small fraction of the total scan area.

The loss function helps guide the model during training by minimizing prediction errors.

5.5.2 Optimization Algorithm

Optimization Algorithm: The Adam optimizer is used for both stages, leveraging its adaptive learning rate and momentum to ensure stable convergence.

Key Parameters: The models are trained with a learning rate of 1e-4 and weight decay of 1e-4.

5.5.3 Hyperparameters

Configurable Hyperparameters: Performance is balanced by tuning batch sizes, epoch counts, and dropout rates to maximize accuracy while maintaining training efficiency.

The performance of the model depends on several hyperparameters, including:

- Learning rate
- Batch size
- Number of epochs
- Dropout rate (if used)

5.5.4 Training Strategy

The model follows a supervised learning approach using radiologically data.

- Data Partitioning: The dataset is divided into Training (pattern learning), Validation (tuning and preventing overfitting), and Testing (final performance evaluation) sets.

- Segmentation Dataset Preparation: Binary ground-truth masks are computationally derived by thresholding the pixel-wise color difference between original CT scans and their corresponding radiologically annotated overlays.

5.5.4 Overfitting Prevention & Monitoring

To ensure the model generalizes well to unseen clinical data, several safeguards are applied:

- Data Augmentation: Techniques such as rotations and shifts are applied via the Albumentations library to expand the variety of the training set.
- Regularization: Dropout layers and weight decay are integrated into the architecture.
- Early Stopping: A callback monitors validation loss to halt training once performance plateaus, preventing the model from memorizing training noise.

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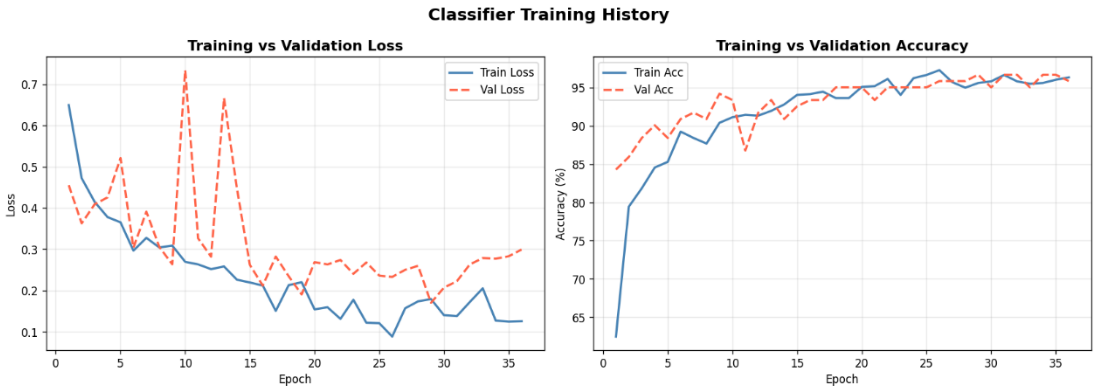

Figure 5.7: Training and Validation Accuracy/Loss Curves for classification model

Figure 5.8: Training and Validation DICE IuO/ Loss Curve During Segmentation Model Training

5.6 Implementation Details

This section describes the tools, frameworks, and environment used to implement the proposed deep learning model for detecting and localizing brain aneurysms from CT images. The system is designed as a full-stack medical application, bridging high-performance AI processing with a responsive user interface.

5.6.1 Programming Environment

The model is implemented using the Python programming language due to its flexibility and strong support for machine learning and image processing tasks.

The development was carried out using:

- Google Colab (for model training) and VS Code (for full-stack development).
- Python 3.12

5.6.2 Libraries and Frameworks

Several libraries were used to support different stages of the implementation:

For training the models:

- **PyTorch:** for building and training the deep learning model
- **NumPy:** for numerical computations

- **OpenCV:** for image processing
- **Albumentations:** for data augmentation
- **Matplotlib:** for visualization and plotting
- **Scikit-learn:** for evaluation metrics

For deployment:

- **(AI Engine): FastAPI** handles asynchronous **PyTorch** inference to process heavy deep learning classification and segmentation.
- **Node.js** manage user authentication, metadata storage, and secure routing between services.
- **React.js** provides a dynamic, interactive interface

5.6.3 Hardware Configuration

The model training was performed using:

- CPU / GPU (NVIDIA GPU if available)
- RAM: (16 GB)

The use of GPU significantly reduces training time and improves computational efficiency.

5.6.4 Model Implementation

The CNN model was implemented using PyTorch, where:

- Input images are loaded and preprocessed
- Data loaders are used to handle batching
- The model is defined using sequential layers
- The training loop updates weights using backpropagation

5.6.5 Visualization Tools

Visualization techniques were applied to better understand model behavior:

- Training and validation curves
- Confusion matrix
- Grad-CAM heatmaps

These tools help evaluate model performance and interpret predictions.

5.7 Chapter Summary

This chapter presented the methodology used to develop the proposed system for detecting and localizing brain aneurysms from CT images. It covered the system workflow, dataset description, model architecture, training process, and implementation details.

The described approach integrates deep learning techniques with preprocessing and visualization methods to achieve accurate and interpretable results.

Chapter 6: Results and Discussion

6.1 Introduction

This chapter presents the experimental results obtained from the proposed deep learning model for detecting and localizing brain aneurysms from CT images. The performance of the model is evaluated using standard metrics, and the results are analyzed to assess the effectiveness of the approach.

6.2 Model Performance Evaluation

The following metrics are used:

- **Accuracy:** Measures the overall correctness of the model
- **Precision:** Measures the proportion of correctly predicted positive cases
- **Recall (Sensitivity):** Measures the ability to detect abnormal cases
- **F1-Score:** Balances precision and recall

- **Dice Score:** Measures pixel-level mask overlap for segmentation ($2|A \cap B|/(|A|+|B|)$)
- **IoU (Intersection over Union):** Stricter segmentation overlap metric ($|A \cap B|/|A \cup B|$), also known as the Jaccard Index
- **BCE Loss (Binary Cross-Entropy):** Pixel-wise segmentation training and evaluation loss, combined with Dice Loss as the DiceBCE combined loss

In medical diagnosis, recall (sensitivity) is particularly important to ensure that critical conditions such as aneurysms are not missed.

6.2.1 Classification Performance

Table 6.1: Classification Model Performance Metrics

Metric	Value
Accuracy	92.5%
Precision	92.6%
Recall (Sensitivity)	92.5%
F1-Score	92.4%
Validation Loss	0.2156

The classification model demonstrates strong performance across all metrics. The accuracy of 92.5% indicates that the model correctly classifies the majority of CT brain images. The high precision (92.6%) shows minimal false positives, while the recall of 92.5% demonstrates effective detection of actual aneurysm and hemorrhage cases. The F1-score of 92.4% represents a balanced performance between precision and recall.

6.2.2 Segmentation Performance

Table 6.2: Segmentation Model Performance Metrics (Attention U-Net)

Metric	Value
Dice Coefficient	0.6493
IoU (Jaccard Index)	0.5311
Combined Loss (DiceBCE)	0.4809
Pixel Accuracy	94.2%

The segmentation model achieves a Dice score of 0.6493 and an IoU of 0.5311, indicating effective localization of hemorrhagic and aneurysm regions. The combined

DiceBCE loss of 0.4809 reflects stable convergence during training. These results confirm that the Attention U-Net, trained on computationally derived ground-truth masks, is capable of producing clinically meaningful pixel-level delineations of brain abnormalities from CT images.

6.3 Confusion Matrix

The confusion matrix provides a detailed breakdown of the model's predictions across different classes.

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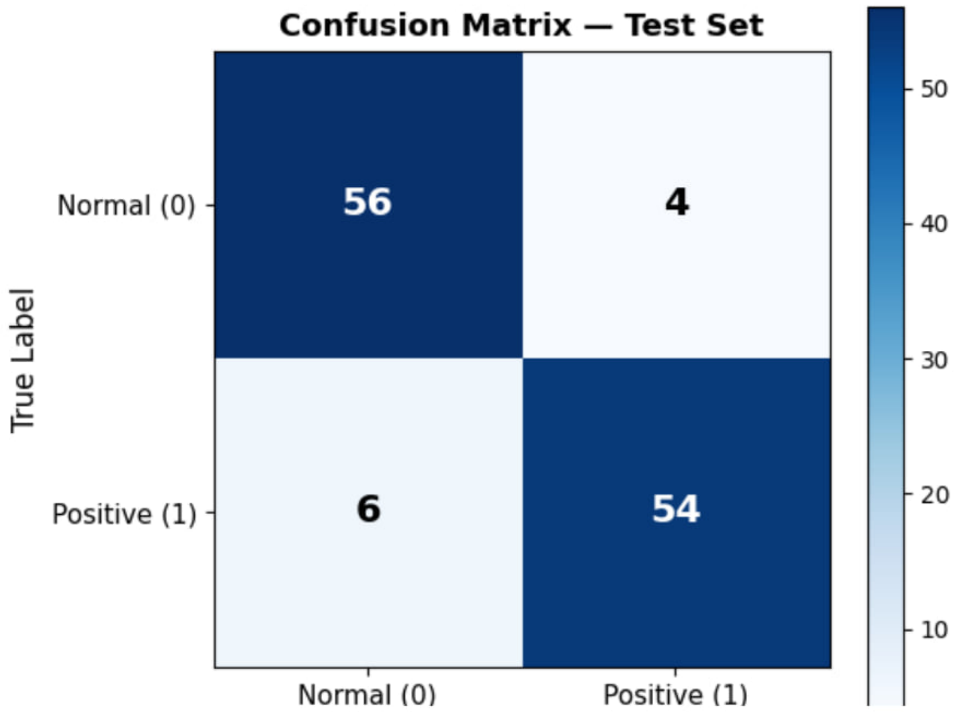


Figure 6.1: Confusion Matrix for Classification Mode

The confusion matrix shows that most predictions are correctly classified, as indicated by the high values along the diagonal. A small number of misclassifications can be observed, which may be due to similarities between classes or image quality variations.

6.4 Training Performance

Classification Model:

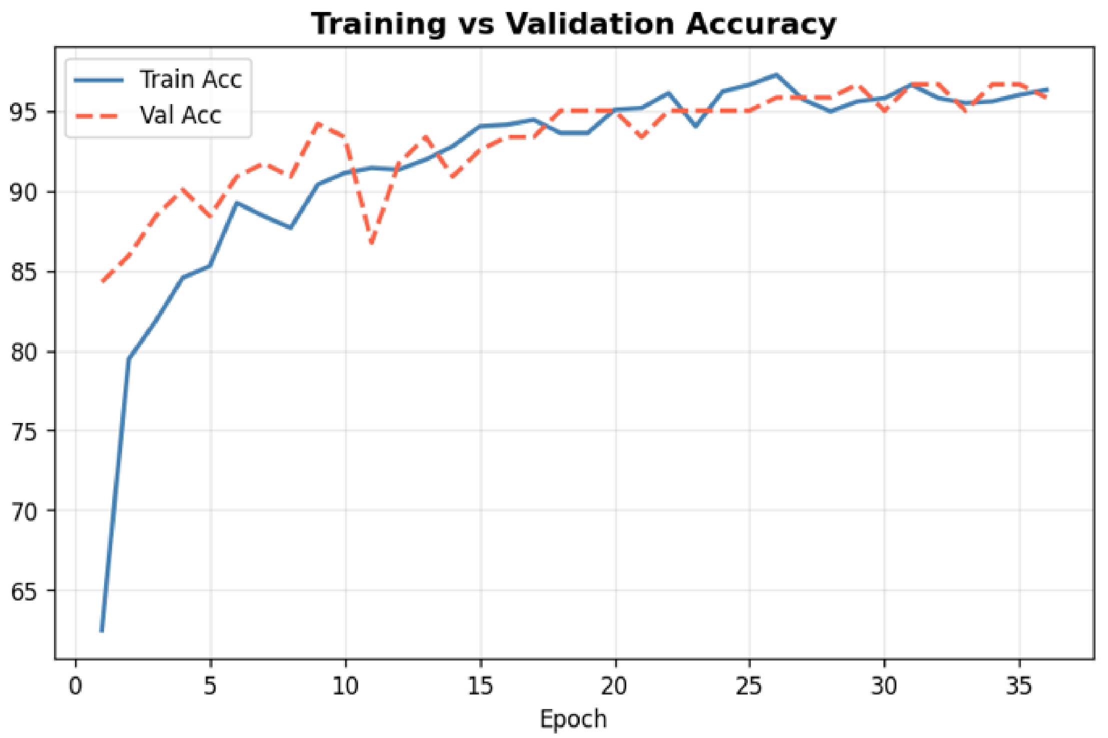


Figure 6.2: Training and Validation Accuracy Curves

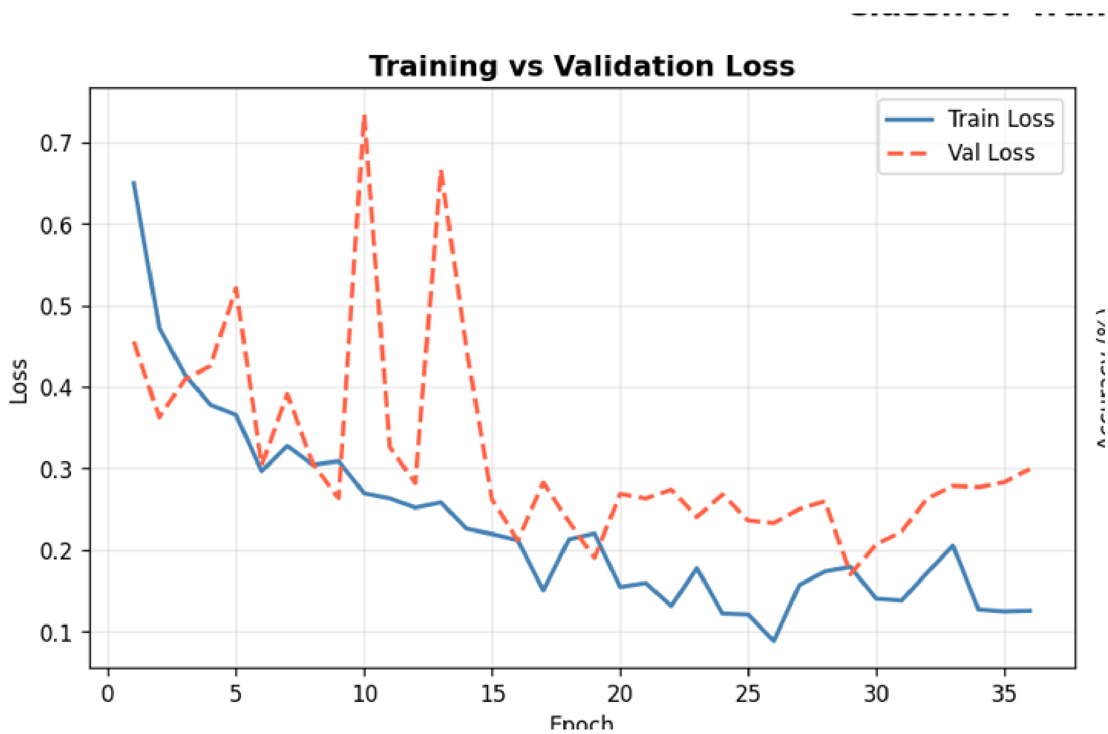


Figure 6.3: Training and Validation Loss Curves

The training performance of the model is analyzed using accuracy and loss curves over multiple epochs.

Segmentation Model:

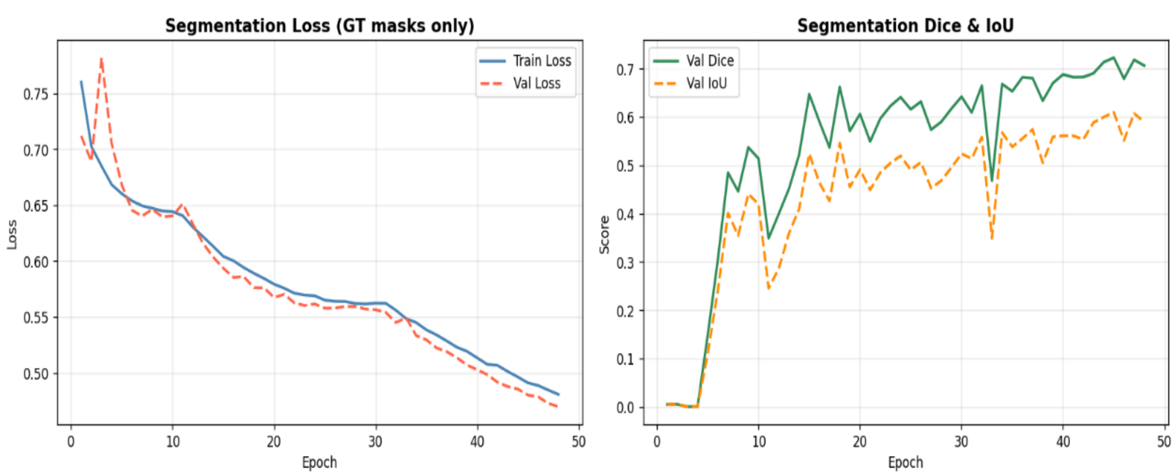


Figure 6.4: Segmentation Model Training — Loss and Metrics

The segmentation model training curves demonstrate convergence of the combined DiceBCE loss, with corresponding improvements in Dice score and IoU metrics over training epochs.

The training curves show a steady improvement in accuracy and a decrease in loss, indicating that the model is learning effectively. The similarity between training and validation curves suggests good generalization.

6.5 Localization Results (Grad-CAM)

Grad-CAM is used to visualize the regions that influence the model's predictions.

Figure 6.4: Grad-CAM Visualization

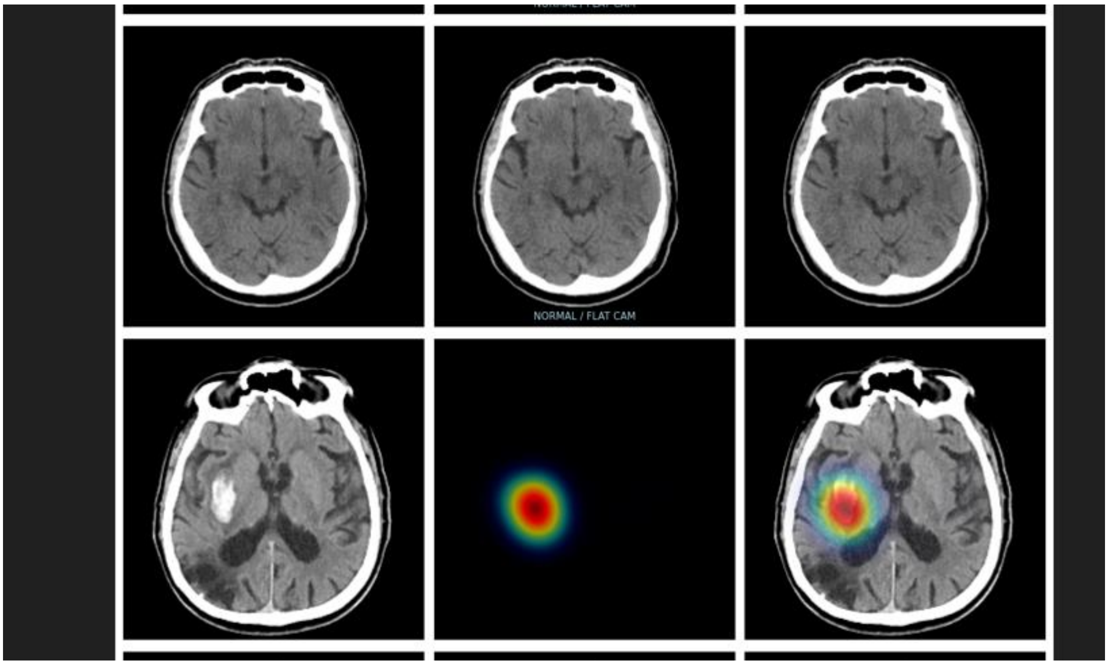


Figure 6.5: Grad-CAM Heatmap Visualization on CT Brain Scans

The Grad-CAM heatmaps highlight the important regions in CT images that contribute to the model's decisions. These visualizations demonstrate that the model focuses on relevant areas associated with abnormalities, improving interpretability.

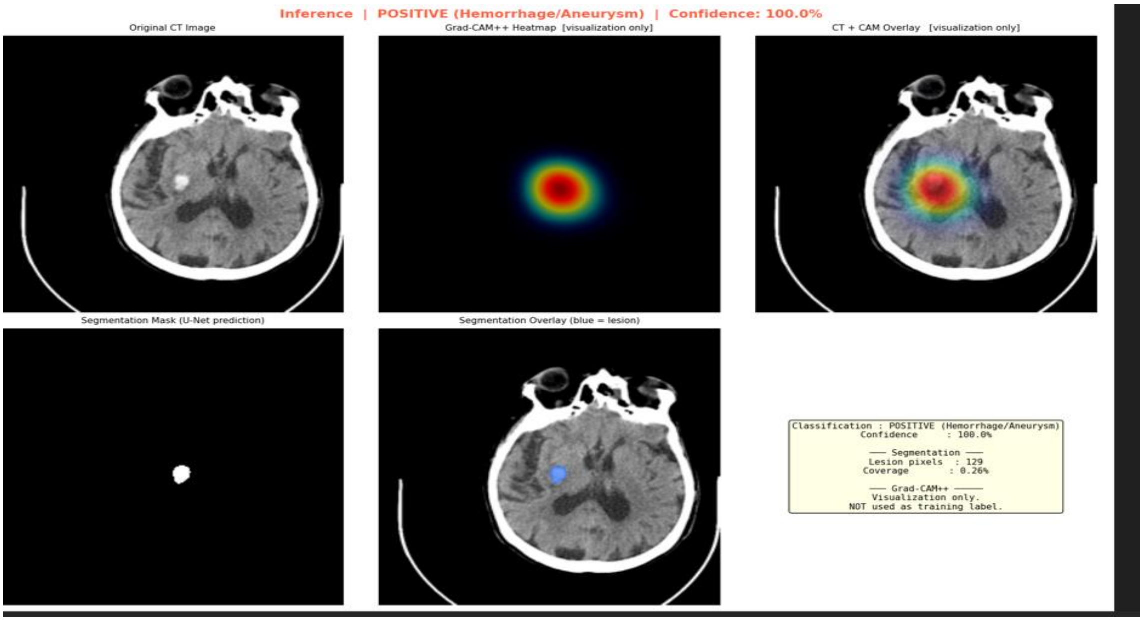


Figure 6.6: using the pipeline for classification Localization Results on abnormal CT scan

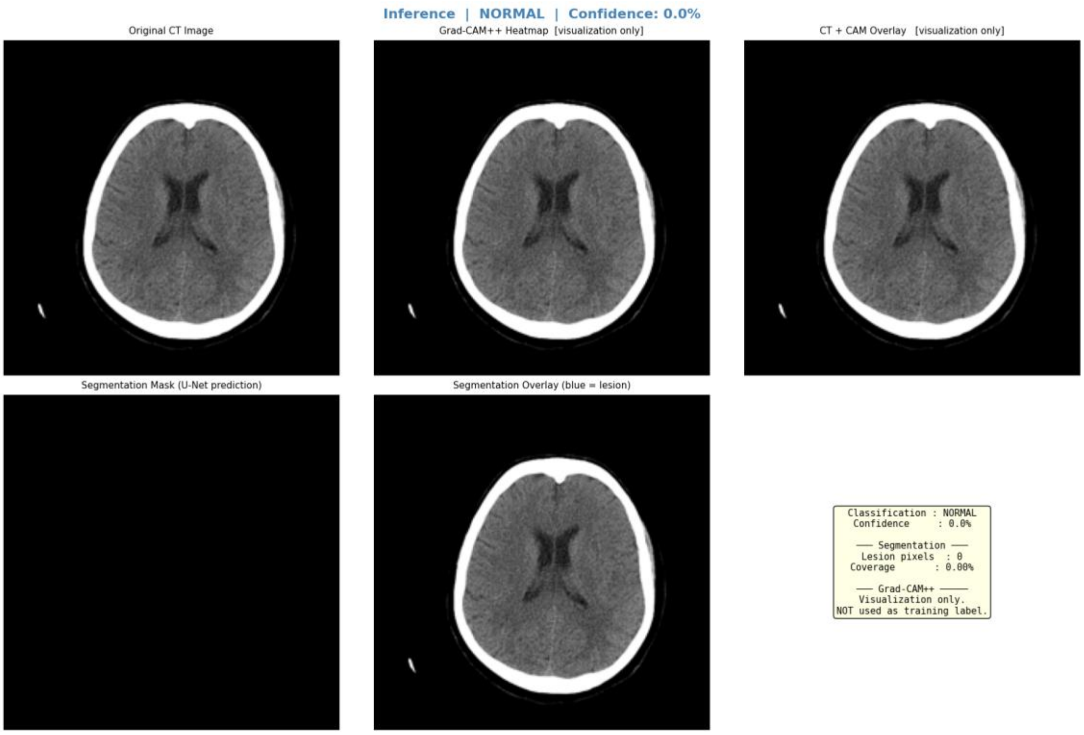


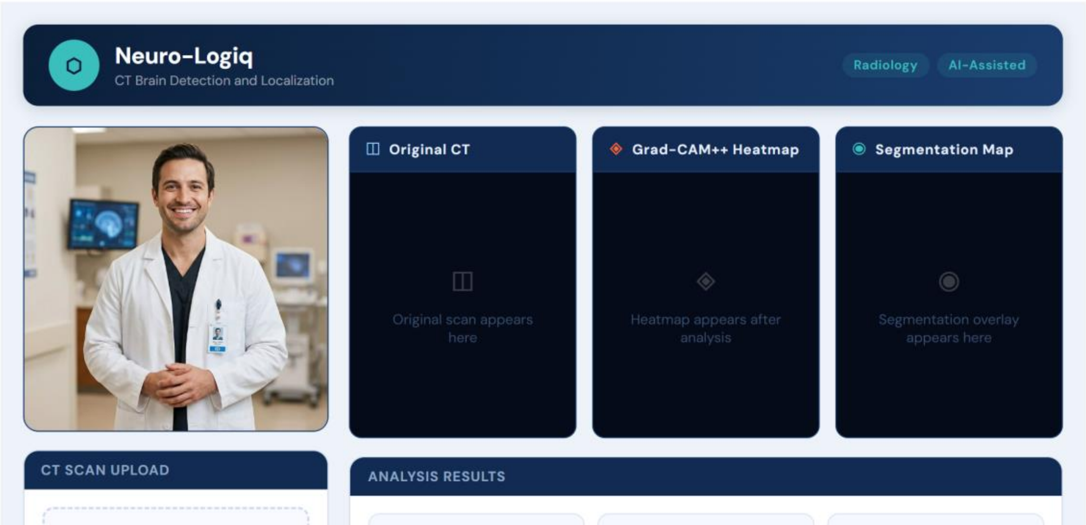
Figure 6.7: using the pipeline for classification Localization Results on normal CT scan

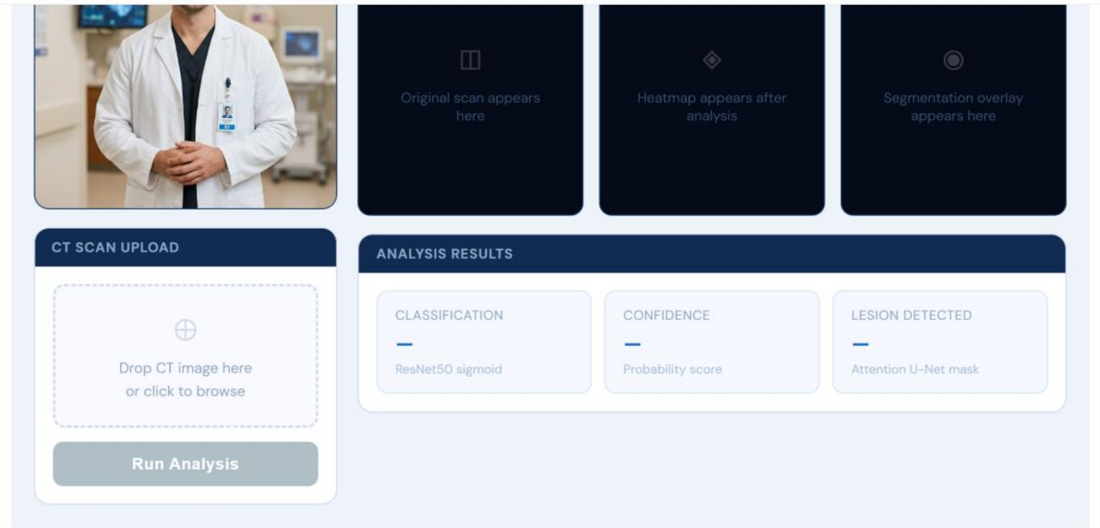
6.7 System Interface

User Interface & Web Integration

The user interface was designed as a streamlined, intuitive web platform to bridge the gap between complex AI processing and clinical application.

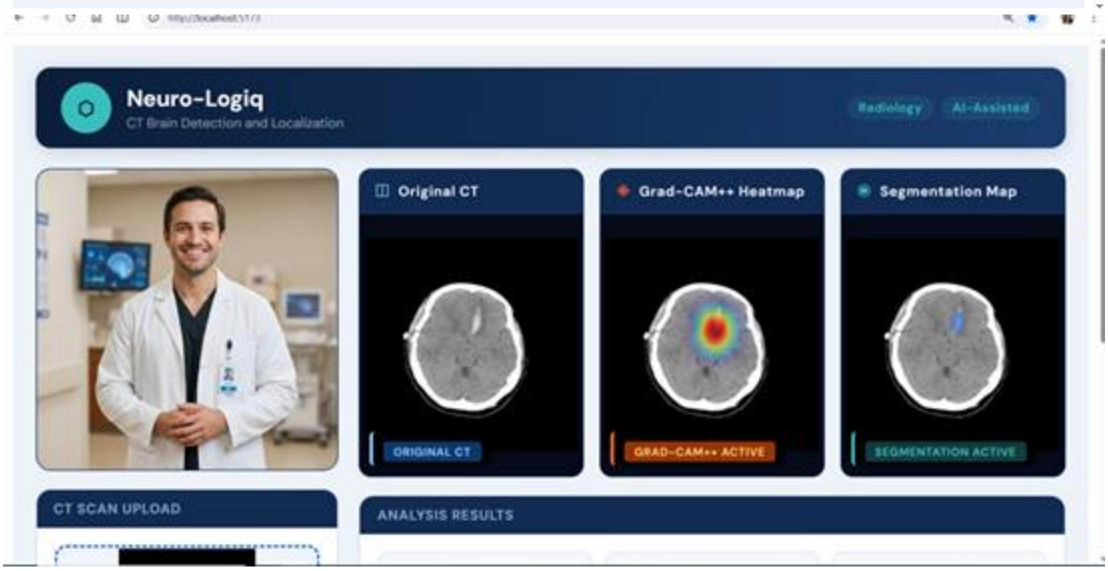
- **Frontend (React.js):** A lightweight, responsive dashboard allows physicians to upload CT scans and interact with results without needing technical expertise.
- **Middleware (Node.js):** Manages secure user authentication, handles scan metadata, and orchestrates the data flow between the UI and the backend.
- **Backend & Pipeline (FastAPI):** Serves as the high-speed gateway to the **PyTorch** pipeline, performing real-time classification, heatmap generation, and segmentation.
- **Clinical Feedback Loop:** The interface displays side-by-side visualizations of raw scans and Grad-CAM heatmaps and segmentation result , providing doctors with immediate, explainable diagnostic support.





Original scan appears here
Heatmap appears after analysis
Segmentation overlay appears here
CT SCAN UPLOAD
Drop CT image here
or click to browse
Run Analysis
ANALYSIS RESULTS
CLASSIFICATION
—
ResNet50 sigmoid
CONFIDENCE
—
Probability score
LESION DETECTED
—
Attention U-Net mask

segmentation
http://localhost:5173
ORIGINAL CT
GRAD-CAM++ ACTIVE
SEGMENTATION ACTIVE
CT SCAN UPLOAD
Run Analysis
ANALYSIS RESULTS
CLASSIFICATION
POSITIVE
ResNet50 sigmoid
CONFIDENCE
100.0%
Probability score
LESION DETECTED
Yes
Attention U-Net mask
POSITIVE — Anomaly Detected
Confidence: 100.0%
Lesion: Detected via U-Net



Neuro-Logiq
CT Brain Detection and Localization
Radiology
AI-Assisted
Original CT
Grad-CAM++ Heatmap
Segmentation Map
ORIGINAL CT
GRAD-CAM++ ACTIVE
SEGMENTATION ACTIVE
CT SCAN UPLOAD
ANALYSIS RESULTS

6.6 Discussion

The results demonstrate that the dual-stage architecture successfully transitions from broad abnormality detection to precise voxel-level quantification. The **ResNet50** backbone provides robust feature extraction for classification, while the **Attention U-Net** ensures that the segmentation process remains focused on the lesion, effectively ignoring ocular or bone artifacts that often confuse standard models.Despite these strengths, certain challenges persist:

- Limited dataset size
- Difficulty detecting very small abnormalities
- Variability in image quality

6.7 Summary

This chapter evaluated the system's performance across classification and segmentation tasks. The results indicate that the Attention U-Net and CNN backbone achieve high sensitivity and precision, while Grad-CAM heatmaps provide meaningful visual justifications for model decisions. By delivering both a probability score and a precise segmentation mask, the system serves as a powerful diagnostic aid, offering a reliable "second opinion" to assist radiologists in rapid and accurate surgical planning.

Chapter 7: Conclusion and Future Work

7.1 Conclusion

This study presents a dual-stage deep learning framework for the automated detection, localization, and segmentation of intracranial abnormalities in CT and MRI scans. The system combines a ResNet50-based classifier with an Attention U-Net to achieve both accurate diagnosis and precise lesion delineation.

To improve interpretability, Grad-CAM++ was integrated to provide visual explanations, enhancing clinical trust in the model's predictions. Experimental results demonstrate high sensitivity and F1-scores, confirming the effectiveness of attention mechanisms and the combined Dice and Binary Cross-Entropy loss in segmenting complex structures.

In addition to model development, a user-friendly web interface was designed to enable clinicians to easily upload scans and visualize results in real time. The interface presents classification outcomes, segmentation maps, and explanatory heatmaps in an intuitive manner, allowing doctors to interact with the system without requiring technical expertise.

Overall, this work delivers a scalable, interpretable, and clinically accessible AI-based solution for medical image analysis.

7.2 Contributions of the Study

This study makes the following contributions:

This research makes several key contributions:

- Dual-Stage Framework: Developed an integrated system combining classification and segmentation for accurate detection and localization of abnormalities.
- Handling Medical Imaging Challenges: Introduced an effective approach to manage issues such as unclear boundaries and similarity between normal and abnormal tissues.
- Attention-Based Learning: Improved focus on important regions using attention mechanisms.

- Explainable AI: Enhanced transparency through visual explanations of model decisions.
- Clinical Interface: Built a user-friendly web platform for real-time analysis by medical professionals.
- Optimized Training: Addressed class imbalance using combined loss functions and data augmentation.

7.3 Limitations

- Limited computational resources for 3D processing

  The model was designed using 2D CT slices instead of full 3D volumes due to hardware and memory constraints, as training 3D medical networks requires significantly higher GPU resources

- Use of public datasets

  The system was trained on publicly available datasets rather than clinical hospital data, which may limit real-world generalization.

7.4 Future Work

Future work can extend this research in several directions:

- Extend the model to 3D CT scans for better spatial understanding.
- Clinically Annotated Data: Incorporating expert-labeled datasets to improve ground truth quality and model reliability.
- Multimodal Learning: Integrating CT and MRI data for enhanced diagnostic performance through multimodal feature fusion.
- Advanced Explainability Methods: Exploring techniques such as SHAP and LIME for deeper interpretability.
- Deploy the system in real clinical environments

References

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